A. Title and Team

Effects of Lighting Augmentation on Industrial Anomaly Detection

Team: Team 1 | Members: Bavishna Ashok Kumar, Chao Zheng, Jim He, Nithya Anantharaman

We study image-level anomaly detection under lighting changes, using CNNs, pretrained features, reconstruction and controlled augmentation.

B. Problem and Research Question

In industrial inspection, lighting changes can make defect-free products appear anomalous. We will test whether brightness and contrast augmentation improves defect detection under changed lighting while preserving performance under regular lighting.

Question: How do additional brightness and contrast augmentation affect EfficientAD image-level anomaly detection on Rice and Walnuts in MVTec AD 2?

H1: Combined brightness and contrast augmentation improves shifted-light AUROC over native EfficientAD. H2: Regular-light AUROC drops by at most 2 percentage points, our practical tolerance. Paired differences and uncertainty will test these hypotheses. Credible evidence, including negative or mixed findings, defines success.

C. Dataset and Evaluation

MVTec AD 2 [1,4] provides industrial inspection images under different lighting. We selected Rice and Walnuts before model evaluation. Inputs are 2448 × 2048 RGB images; targets are normal/anomalous image labels, with public pixel masks for error analysis.

Category	Train normal	Validation normal	Public test normal / anomaly
Rice	313	35	42 / 90
Walnuts	432	48	60 / 90

Dataset access was confirmed on October 25, 2026: 8,004 images and 705 masks (32.74 GB) across eight categories. The planned subset has 1,110 images and 180 masks. Pinned mirror files [6,7] passed integrity, count and format checks. Private labels remain server-held. CC BY-NC-SA 4.0 applies; no new annotation or human-data collection is needed.

Preserve official splits. Fit preprocessing on training normals; use normal validation for settings, calibration and thresholds. Freeze choices before testing. Group related scene views in uncertainty estimates. Record pretrained-data provenance and known evaluation overlap. Conclusions apply to this public subset.

Primary metric: image AUROC pooled over the five nonregular lighting conditions within each category, then averaged equally across categories. AUROC measures normal/anomalous score ranking without a decision threshold. Also report regular-light and per-light AUROC, AUPRC, recall and false-positive rate. Set each operating threshold to the 95th percentile of normal validation image scores.

D. Related Work and Technical Approach

MVTec AD 2 [1] (2026) motivates native lighting-shift evaluation. EfficientAD [2] (2024) supplies a compact teacher–student and reconstruction baseline. SingleNet [3] (2025) supports efficient distillation and texture-error analysis; we will not implement its Fourier architecture.

Reference baseline: a small convolutional autoencoder trained on normal tiles. Strong baseline: Anomalib EfficientAD-S [5], with a frozen pretrained Small Patch Description Network teacher [9] and Imagenette auxiliary data [8]. We reuse these components and implement the pipeline, reference model, augmentation and evaluation. Both methods use native-resolution 256 × 256 tiles and maximum anomaly-map scoring. Tiling retains small defects but changes global context and latency; published full-image scores are not directly comparable.

E. Experimental Plan

Main comparison: S11 − S00. Brightness ablation: S11 − S01; contrast ablation: S11 − S10. Enabled factors are sampled independently from Uniform(0.8, 1.2), with one shared 50% application gate and brightness before contrast. Evaluation receives no added transforms.

Setting	Extra brightness	Extra contrast	Purpose
S00	Off	Off	Native EfficientAD baseline
S10	On	Off	Brightness component
S01	Off	On	Contrast component
S11	On	On	Combined intervention

Across EfficientAD settings, hold splits, tiles, scoring, pretrained initialization, native losses/augmentation, auxiliary data, optimization, training steps and calibration fixed. Pair seeds 11, 22 and 33; use separate random streams for extra augmentation. EfficientAD uses 40,000 steps, batch size 1 and the implementation's learning-rate schedule. The reference uses 10,000 steps per seed. Full scope totals 24 EfficientAD and 6 reference fits. Fix final-step checkpoints and evaluation rules before testing.

Report regular, underexposed, overexposed, shift_1, shift_2 and shift_3 results, every seed, and mean and standard deviation. A paired scene-cluster bootstrap estimates test-sample uncertainty while keeping lighting views together. The 282 test images represent 47 physical scenes; small scene counts limit significance claims.

Report hardware/software versions, parameters, memory and training time. Measure complete-image inference after warm-up and device synchronization, separating preprocessing and data transfer. Present metric tables and masked error examples, including false positives, false negatives and limitations. Test results will not change models or thresholds.

F. Feasibility and Team Plan

Available compute is a 24 GB computer using PyTorch. Provisional budget: 48 training hours and 8 evaluation hours. A 1,000-step pilot will measure compatibility, memory and runtime before full training; the Rice-only fallback applies if the forecast exceeds the budget.

Milestone	Evidence
Weeks 1–2	Data audit, source manifest, protocol and team agreement; compute pilot
Week 3	Runnable reference baseline, split manifest, dataset card
Week 4	Strong baseline and normal-validation diagnostics; 2–3-page progress report
Week 5	Complete controlled runs and record configurations
Week 6	Freeze protocol; final test evaluation, error and resource analysis
Weeks 7–8	Cross-member reproduction, contribution statements and peer assessments; final report, code and presentation by Dec 16, 2026

Team member	Substantive technical responsibility
[Team member 1]	**Data and reference model.** Verify the dataset and maintain the split and scene records. Build the shared image loader and tiling process. Train the reference autoencoder for three seeds per category and document preprocessing. **Deliverables:** Shared data loader, split manifest, autoencoder code and saved models.
[Team member 2]	**Baseline model.** Set up EfficientAD-S, its pretrained teacher and auxiliary data. Implement standard training and calibration, then train the original setting (S00) for three seeds per category. **Deliverables:** Reusable EfficientAD training code, baseline saved models and training logs.
[Team member 3]	**Lighting experiments.** Use the baseline code to implement brightness only (S10), contrast only (S01) and both (S11). Train these settings using the planned seeds and fixed controls. Record configurations and training time. **Deliverables:** Augmentation code, experiment configurations, saved models and run logs.
[Team member 4]	**Evaluation and analysis.** Evaluate all saved models using the fixed protocol and validation-based thresholds. Calculate detection metrics and scene-based uncertainty, measure inference time, and inspect errors across lighting conditions. **Deliverables:** Evaluation code, metric tables, comparison figures and findings.

Each member will code, analyze and write their part, cross-review another member and explain the full system and one experiment. A non-author will reproduce an evaluation from the README and record commands, configuration, result and discrepancies. Minimum scope is Rice (four settings, three seeds, reference; 15 fits); full scope adds Walnuts (30). Tiling cost is the main risk; few scenes limit conclusions. Discuss budget-driven scope reductions with the instructor before testing. Private-server evaluation is a stretch goal.

References

[1] L. Heckler-Kram, J.-H. Neudeck, U. Scheler, R. König, and C. Steger. The MVTec AD 2 Dataset: Advanced Scenarios for Unsupervised Anomaly Detection. International Journal of Computer Vision, 134, article 175, March 2026. Source

[2] K. Batzner, L. Heckler, and R. König. EfficientAD: Accurate Visual Anomaly Detection at Millisecond-Level Latencies. WACV 2024, pp. 128–138. Source

[3] S. Thomine and H. Snoussi. Single-Layer Distillation with Fourier Convolutions for Texture Anomaly Detection. WACV 2025, pp. 8962–8971. Source

[4] MVTec Software. MVTec AD 2 dataset, download information, and CC BY-NC-SA 4.0 license. Accessed October 25, 2026. Source

[5] Open Edge Platform. Anomalib EfficientAD implementation. Source version reviewed October 25, 2026. Source

[6] Y. Jiang. Anomagic researcher repository linking the AnomVerse MVTec AD 2 mirror. Selected original image folders only; no added crops, overlays, or generated metadata. Source

[7] AnomVerse. MVTec AD 2 data mirror, fixed source version accessed October 25, 2026. Source

[8] fast.ai. Imagenette, a ten-class subset of ImageNet used as auxiliary penalty data by the selected EfficientAD implementation. Dataset URL accessible October 25, 2026. Source

[9] Open Edge Platform. EfficientAD pretrained_teacher_small.pth, efficientad_pretrained_weights release. Asset URL accessible October 25, 2026. Source

Assistance disclosure

OpenAI Codex assisted literature discovery, experimental-design suggestions, substantive drafting and editing. Publication details were checked against publisher/proceedings pages; implementation claims against pinned Anomalib source. Dataset counts and formats were checked against the publication, with file integrity recorded in local audits. No model training or performance results are claimed.